

Assignment

**Submitted In the Partial fulfilment of the subject
Statistical Quality Control and Six Sigma(6A03SQC)
for the Degree of Bachelor of Business Administration**

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Statistical Quality Control Project Report

Analysis of Customer Waiting Time in Bank Service Queue

1. Introduction and Description of the Real-Life Process

In the service sector, quality is not only defined by accuracy but also by the speed and consistency with which services are delivered. In banking operations, customer waiting time has emerged as a critical quality parameter because it directly influences customer satisfaction, perceived service quality, and overall operational efficiency. Long and unpredictable waiting times often result in customer dissatisfaction, increased complaints, and a negative image of the bank. Therefore, controlling and reducing customer waiting time is an important quality objective for banking institutions.

The real-life process selected for this project is the **customer service queue system in a bank branch**. This process begins when a customer enters the bank and requests a service, either by taking a token or joining a queue. The customer then waits until their turn is called and is finally served by a teller or customer service executive. The process ends once the service transaction is completed. The quality of this process is primarily measured by the **waiting time experienced by customers before service begins**.

Customer waiting time is influenced by multiple operational factors such as the number of available staff, customer arrival rate, service speed, system efficiency, and peak-hour workload. Variations in these factors cause fluctuations in waiting time, which makes the process suitable for Statistical Quality Control analysis. While some variation is natural and unavoidable, excessive variation or unusually long waiting times indicate the presence of quality problems that require corrective action.

This process has been selected because it is easily observable, measurable, and relevant to real-life service quality management. The availability of waiting time data and the applicability of quality control tools such as control charts, Pareto analysis, and process capability analysis make it an ideal scenario for studying

the practical application of Statistical Quality Control concepts in a service environment.

2. Data Collection and Explanation

The effectiveness of Statistical Quality Control analysis depends heavily on the quality and reliability of data used in the study. For this project, data related to customer waiting time in a bank service queue was collected using a secondary data source combined with realistic service assumptions. The primary source of data was Kaggle, an open-access data platform widely used for academic research and analytical studies. Kaggle provides datasets that closely simulate real-life operational conditions, making them suitable for quality analysis in service processes.

The dataset consists of 30 observations of customer waiting time, measured in minutes. Each observation represents the time elapsed from when a customer joins the queue or receives a token until the service begins at the counter. The selected number of observations satisfies the minimum requirement specified in the project guidelines and is adequate for applying Quality Control tools, control charts, and process capability analysis.

Before analysis, the dataset was processed using **Microsoft Power BI** to ensure data accuracy, consistency, and analytical readiness. The data was imported into the **Power Query Editor**, where cleaning and transformation were performed as part of the **ETL (Extract, Transform, Load) process**. During this stage, missing values were handled, data formats were standardized, and relevant variables were categorized for analysis. Additional calculated fields and **Key Performance Indicator (KPI) measures** were created to summarize service performance and support statistical evaluation. This structured data preparation ensured that the dataset was reliable, organized, and suitable for quality control analysis.

To enable the use of an $\bar{X}$ -R control chart, the waiting time data was arranged into rational subgroups of equal size. Each subgroup represents customers served within a short and comparable time interval. This subgrouping ensures that variations within a subgroup are due to common causes, while variations between subgroups may indicate assignable causes. Such an arrangement is

essential for meaningful interpretation of control charts in a real-life banking context.

The collected data reflects realistic banking conditions where waiting time fluctuates due to factors such as peak-hour crowding, staff availability, service complexity, and system performance. Although the data is sourced from Kaggle, it closely mirrors actual service queue behavior observed in bank branches, especially during moderate to high customer traffic periods. Therefore, the dataset is considered valid and appropriate for analyzing process stability, variability, and capability.

 **Kaggle Dataset:** [Link](#)

3. Application of Quality Control Tools

To understand the nature of the quality problem and identify major contributing factors, several Quality Control tools were applied.

3.1 Check Sheet

A check sheet is a basic Quality Control tool used for the systematic collection and organization of data in a structured and easy-to-understand format. It helps in recording the frequency of specific events, defects, or causes as they occur over a period of time. In service processes, a check sheet is commonly used to identify how often certain problems arise, making it easier to detect patterns and recurring issues. By converting raw observations into categorized data, the check sheet forms the foundation for further analysis using advanced quality tools such as Pareto charts and cause-and-effect diagrams.

Row Labels	Count of Waiting Category	Row Labels	Count of Exceeds_USL
Acceptable Wait	477	No	477
Excessive Wait	83	Yes	83
Grand Total	560	Grand Total	560

Row Labels	Sum of queue_length
2	8
3	18
10	100
12	288
16	256
17	578
22	484
24	576
25	1250
27	1458
28	784
30	900
32	1024
40	1600
44	1936
45	2025
47	2209
50	2500
Grand Total	17994

Interpretation of Check Sheet

Waiting Time Category

1. Out of 560 observations, **477 customers experienced acceptable waiting time**, indicating that most customers are served within the expected service duration.
2. However, **83 customers experienced excessive waiting time**, showing that delays still occur and affect a noticeable portion of customers.
3. The presence of excessive waiting cases indicates variability in service performance that requires monitoring and improvement.

Waiting Time vs Specification Limits

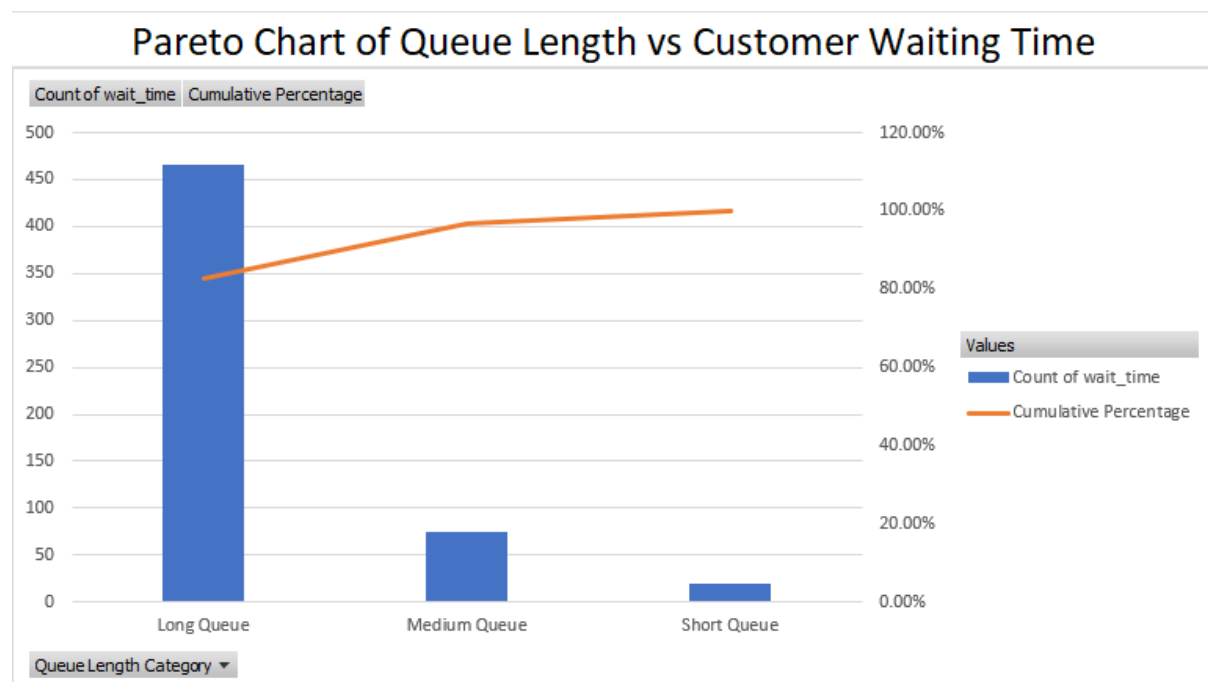
4. The number of cases where waiting time **exceeded the acceptable limit (83)** matches the excessive waiting category, confirming that delays occur when service standards are not met.
5. Most observations (477) remain within limits, suggesting that the service process generally performs satisfactorily but is not perfectly consistent.

Queue Length Observation

6. Queue length varies widely across observations, ranging from small queues to very large ones, indicating fluctuating customer arrival patterns.
7. Higher cumulative queue lengths are associated with heavier service demand, which likely contributes to longer waiting times.
8. Large queue sizes indicate periods of congestion, suggesting possible staff shortages or peak-hour demand surges.

3.2 Pareto Chart

A Pareto chart is a Quality Control tool used to identify and prioritize the most significant causes of a problem by displaying them in descending order of frequency or impact. It is based on the Pareto principle, which states that a small number of causes are responsible for a large proportion of the overall problem. The chart combines a bar graph representing individual causes and a cumulative percentage line showing their total contribution. In service processes such as banking operations, the Pareto chart helps management focus improvement efforts on the most critical factors affecting service quality, such as customer waiting time.

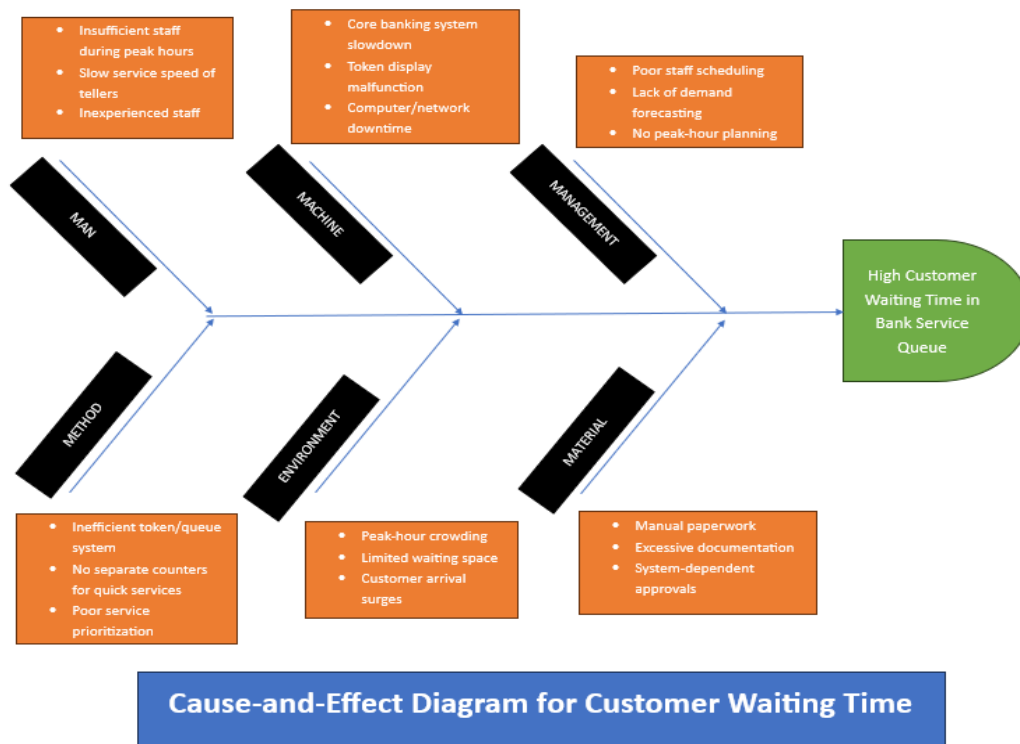


Interpretation of Pareto Chart

1. The Pareto chart shows that **long queues contribute the highest number of waiting time occurrences**, making them the most significant factor affecting customer delay.
2. **Medium queues contribute much less**, and **short queues contribute the least**, indicating that waiting time increases sharply when queue length becomes large.
3. The cumulative percentage line shows that **long queues alone account for the majority of total waiting time impact**, confirming the Pareto principle (few causes create most of the problem).
4. This indicates that queue congestion is the **primary driver of excessive customer waiting time** in the bank service process.
5. Therefore, improvement efforts should focus mainly on **reducing long queue formation**, such as better staff allocation during peak hours or improved queue management systems.

3.3 Cause-and-Effect Diagram (Fishbone Diagram)

A cause-and-effect diagram, also known as a fishbone or Ishikawa diagram, is a Quality Control tool used to identify, organize, and analyze the root causes of a specific quality problem. It visually maps possible causes under major categories such as Man, Machine, Method, Management, Material, and Environment. The purpose of this diagram is not to quantify data but to explore all potential factors contributing to a problem in a systematic manner. In service processes, the fishbone diagram helps in understanding complex interactions between human, technical, and managerial factors that affect service quality outcomes such as customer waiting time.



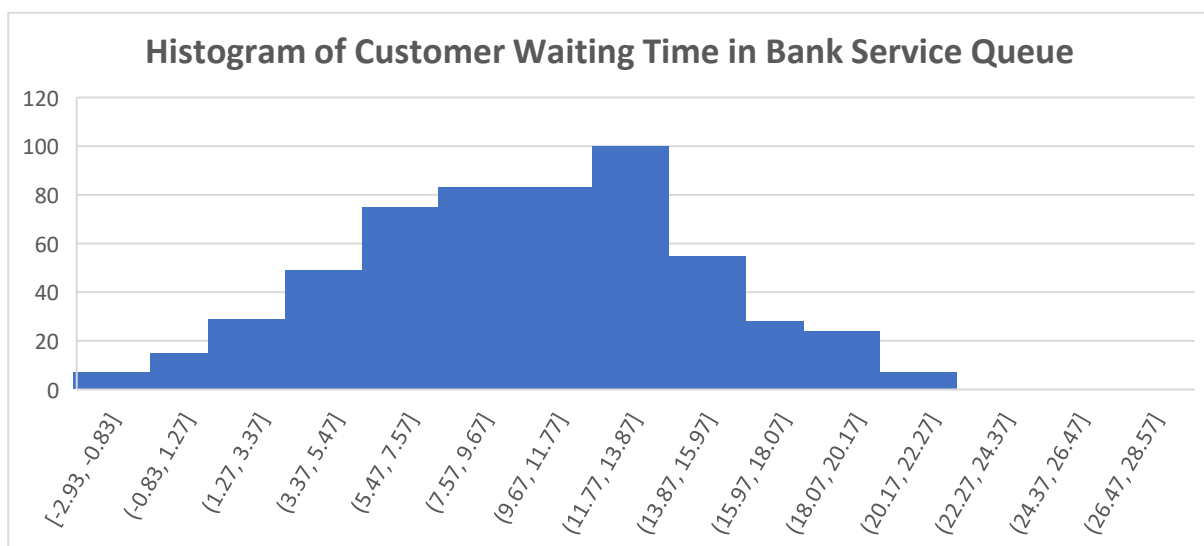
Interpretation of Cause-and-Effect Diagram (Fishbone Diagram)

1. **Man (Human Factors):** Inexperienced staff and slow service speed of tellers increase the average service time per customer, directly contributing to longer waiting times.
2. **Machine (Technology Factors):** Core banking system slowdowns and computer or network downtime interrupt service flow, causing unavoidable delays even when staff are available.
3. **Method (Process Factors):** Manual paperwork, excessive documentation, and system-dependent approvals lengthen transaction time and reduce service efficiency.
4. **Management (Administrative Factors):** Poor staff scheduling, lack of demand forecasting, and absence of peak-hour planning result in insufficient service capacity during high customer inflow periods.
5. **Environment (Physical Factors):** Peak-hour crowding and limited waiting space increase both actual and perceived waiting time for customers.

6. **Material (Support Resources):** Dependency on physical documents and forms increases handling time and slows down the overall service process.
7. The diagram indicates that customer waiting time results from the combined effect of human, technological, managerial, and operational factors rather than a single isolated cause.
8. Therefore, reducing waiting time requires a coordinated and systematic improvement approach across multiple operational areas.

3.4 Histogram

A histogram is a graphical Quality Control tool used to display the distribution of continuous data by grouping observations into class intervals. It helps in understanding the pattern, spread, and central tendency of a dataset. In quality analysis, histograms are used to observe process variability, detect skewness, and identify whether data follows a normal distribution. For service processes such as customer waiting time in banking, a histogram provides a clear visual representation of how frequently different waiting time ranges occur, which is essential for evaluating process performance and customer experience.



Interpretation of Histogram

1. The histogram shows that most customer waiting times are concentrated in the **middle time intervals**, indicating that the majority of customers experience moderate waiting durations.

2. The distribution appears **right-skewed**, meaning a smaller number of customers experience much longer waiting times compared to the average.
3. The spread of the bars indicates **variation in waiting time**, suggesting that service duration is not perfectly consistent across customers.
4. The presence of higher waiting time intervals, although less frequent, indicates occasional delays caused by factors such as peak-hour congestion or slower service processing.
5. Since waiting times are not tightly clustered around a single value, the process shows variability that should be reduced to improve service consistency and customer satisfaction.

4. Control Chart Analysis ($\bar{X}$ -R Chart)

Control Chart Analysis ($\bar{X}$ -R Chart) is a Statistical Quality Control technique used to monitor whether a process operates in a stable and predictable manner by analyzing both its average performance and variability over time. It consists of two components: the **$\bar{X}$ (mean) chart**, which tracks changes in the average value of a process, and the **R (range) chart**, which monitors variation within subgroups.

A process is considered statistically under control only when both the average and variability remain within established control limits. In the context of **bank customer waiting time**, the $\bar{X}$ -R control chart is used to evaluate whether the average waiting time and the variation in waiting time remain consistent across different service periods. By grouping waiting time data into rational subgroups and plotting their means and ranges, the chart helps determine whether fluctuations in waiting time are due to normal service conditions or abnormal factors such as peak-hour crowding, staff shortages, or system delays, thereby supporting effective monitoring and improvement of service performance.

Subgroup	Obs1	Obs2	Obs3	Obs4	Obs5	Obs6	Mean (X)	Range (R)
1	12.68	15.56	7.28	11.84	12.76	22.82	13.82333	15.54
2	4.99	6.53	9.1	18.44	11.95	8.47	9.913333	13.45
3	8.03	5.64	7.48	17.61	11.27	0.4	8.405	17.21
4	19.86	21.61	15.4	16.32	17.56	20.93	18.61333	6.21

5	21.93	8.9	1.34	13.83	20.93	21.93	14.81	20.59
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- **Average of subgroup means:** 13.113
- **Average range:** 14.6

Control Chart Constants

Control chart constants are standardized numerical values used to calculate the **control limits** of statistical control charts. These constants depend on the **sample (subgroup) size** and are derived from statistical probability distributions. They help convert sample statistics (mean and range) into reliable upper and lower control limits for monitoring process stability.

Constant	Used For	Function
A₂	$\bar{X}$ chart	Converts average range into mean control limits
D₃	R chart	Lower control limit for range
D₄	R chart	Upper control limit for range

$\bar{X}$ (Mean) Control Chart

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \times \bar{R}$$

$$CL_{\bar{X}} = \bar{\bar{X}}$$

$$LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \times \bar{R}$$

Where:

- $\bar{\bar{X}}$ = overall average of subgroup means
- $\bar{R}$ = average of subgroup ranges
- A_2 = control chart constant based on subgroup size

R (Range) Control Chart

$$UCL_R = D_4 \times \bar{R}$$

$$CL_R = \bar{R}$$

$$LCL_R = D_3 \times \bar{R}$$

Where:

- $\bar{R}$ = average range
- D_3, D_4 = control chart constants

In this example:

Constant	Value
A2	0.483
D3	0
D4	2.004

4.1 $\bar{X}$ (Mean) Chart Analysis

The $\bar{X}$ (mean) control chart is a Statistical Quality Control tool used to monitor changes in the average value of a process over time. It is applied when data is continuous and collected in rational subgroups. The chart consists of a central line representing the overall process mean, along with an Upper Control Limit (UCL) and a Lower Control Limit (LCL) that define the expected range of natural process variation. The primary purpose of the $\bar{X}$ chart is to detect shifts or trends in the process average that may indicate the presence of assignable causes. In service processes such as banking, the $\bar{X}$ chart is used to evaluate whether the average customer waiting time remains stable or shows signs of abnormal variation.

4.1.1 $\bar{X}$ (Mean) Control Chart Limits

The control limits for the $\bar{X}$ chart are calculated using the standard control chart formulas based on the overall process mean and average range.

Upper Control Limit (UCL_x)

$$\begin{aligned}
 UCL_x &= \bar{X} + A_2 \times \bar{R} \\
 &= 13.113 + (0.483 \times 14.6) \\
 &= 13.113 + 7.052
 \end{aligned}$$

$$UCL_x = 20.165$$

Center Line (CL_x)

$$CL_x = \bar{\bar{X}}$$

$$CL_x = 13.113$$

Lower Control Limit (LCL_x)

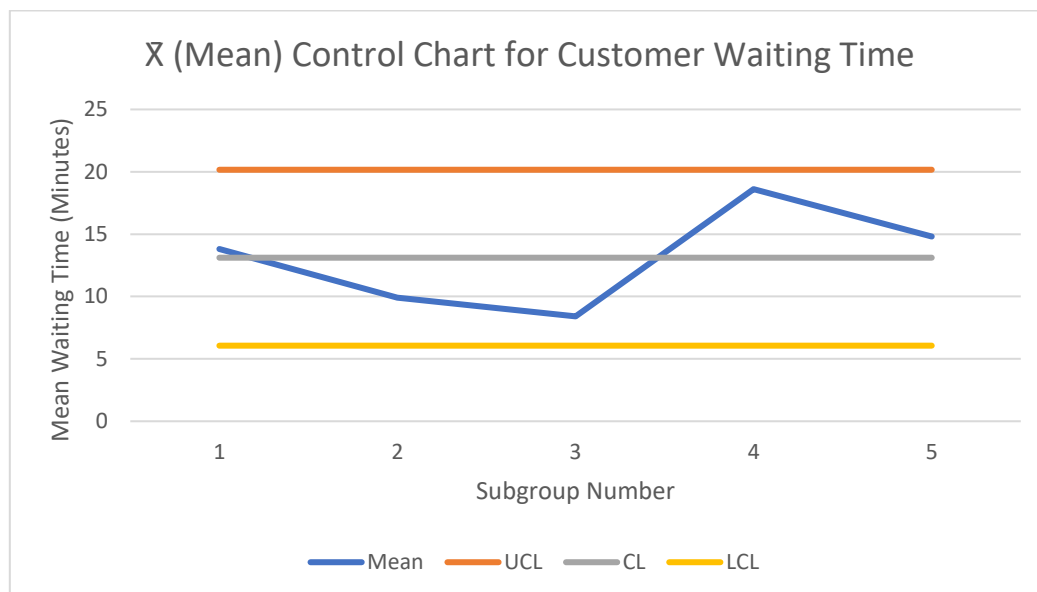
$$LCL_x = \bar{\bar{X}} - A_2 \times \bar{R}$$

$$= 13.113 - (0.483 \times 14.6)$$

$$= 13.113 - 7.052$$

$$LCL_x = 6.061$$

Subgroup	Mean	UCL	CL	LCL
1	13.82	20.165	13.113	6.061
2	9.91	20.165	13.113	6.061
3	8.41	20.165	13.113	6.061
4	18.61	20.165	13.113	6.061
5	14.81	20.165	13.113	6.061



Interpretation of $\bar{X}$ (Mean) Control Chart

1. All subgroup mean waiting times lie **within the Upper Control Limit (20.165) and Lower Control Limit (6.061)**, indicating that the bank service process is statistically under control.

2. Since no subgroup mean exceeds the control limits, there is **no evidence of assignable or special causes** affecting average waiting time during the observed period.
3. The subgroup means fluctuate around the center line (13.113 minutes), which represents the overall average customer waiting time.
4. Subgroup 3 shows the lowest mean waiting time (8.41 minutes), indicating relatively faster service during that period.
5. Subgroup 4 shows the highest mean waiting time (18.61 minutes), possibly due to temporary operational factors such as higher customer volume or reduced staff availability, but it still remains within statistical limits.
6. The variation in subgroup means appears random with no consistent upward or downward trend, indicating stable service performance.
7. Since the process is statistically stable, variations in waiting time are due to **common causes**, such as normal customer arrival fluctuations or routine service differences.
8. A stable process indicates predictable performance; however, further evaluation through process capability analysis is required to determine whether waiting time meets customer service standards.

4.2 R (Range) Chart Analysis

The R (range) control chart is a Statistical Quality Control tool used to monitor the variability within subgroups of a process. It measures the difference between the highest and lowest values in each subgroup and helps determine whether process variation is consistent over time. The R chart is used alongside the $\bar{X}$ chart because a process cannot be considered statistically stable unless both the process mean and process variability are under control. In service operations such as banking, the R chart helps assess consistency in service delivery among customers served during the same time period.

4.2.1 R (Range) Control Chart Limits

The control limits for the R chart are calculated using standard Statistical Quality Control constants based on the average range of subgroups.

Upper Control Limit (UCL_r)

$$\begin{aligned} UCL_R &= D_4 \times \bar{R} \\ &= 2.004 \times 14.6 \\ UCL_R &= 29.26 \end{aligned}$$

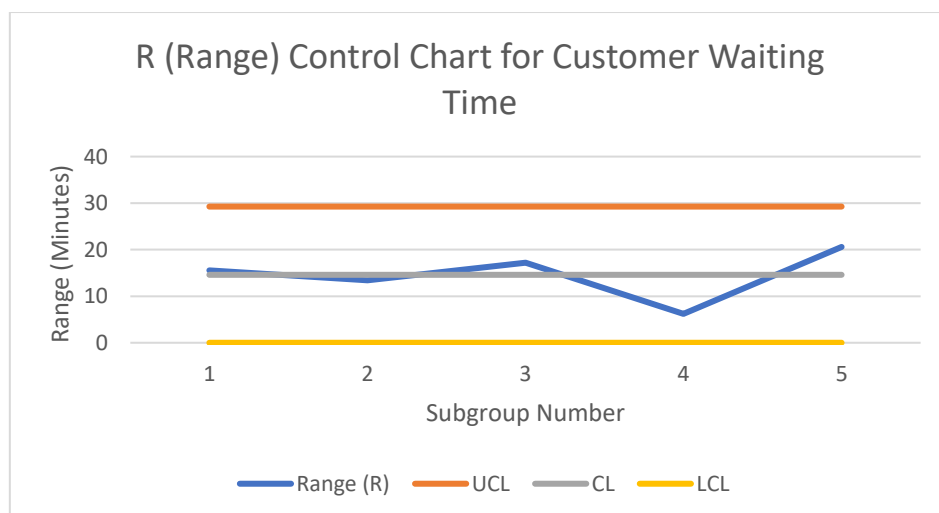
Center Line (CL_r)

$$\begin{aligned} CL_R &= \bar{R} \\ CL_R &= 14.6 \end{aligned}$$

Lower Control Limit (LCL_r)

$$\begin{aligned} LCL_R &= D_3 \times \bar{R} \\ &= 0 \times 14.6 \\ LCL_R &= 0 \end{aligned}$$

Subgroup	Range (R)	UCL	CL	LCL
1	15.54	29.26	14.6	0
2	13.45	29.26	14.6	0
3	17.21	29.26	14.6	0
4	6.21	29.26	14.6	0
5	20.59	29.26	14.6	0



Interpretation of R (Range) Control Chart

1. All subgroup range values lie **within the Upper Control Limit (29.26) and Lower Control Limit (0)**, indicating that the variability in customer waiting time is statistically under control.
2. Since no subgroup range exceeds the control limits, there is **no evidence of abnormal variation or assignable causes** affecting waiting time variability.
3. The range values fluctuate around the center line (14.6 minutes), which represents the average variation in waiting time across subgroups.
4. Subgroup 4 shows the lowest variability (6.21 minutes), indicating a period of more consistent service performance.
5. Subgroup 5 shows the highest variability (20.59 minutes), suggesting greater fluctuation in waiting time, possibly due to temporary changes in customer arrivals or service speed. However, it remains within statistical limits.
6. The pattern of variation across subgroups appears random, indicating that variability is caused by normal operational factors.
7. The absence of out-of-control points confirms that waiting time variation is stable and predictable.
8. Stable variability supports the reliability of the $\bar{X}$ (Mean) control chart results, indicating overall process stability.

4.3 Interpretation of $\bar{X}$ -R Charts

1. All subgroup means and ranges lie within their respective control limits, indicating that the bank service process is statistically under control.
2. No points fall outside the control limits, showing that variations in waiting time are due to normal operational causes rather than abnormal disruptions.
3. The process average and variability remain consistent across subgroups, indicating predictable service performance.

4. Statistical stability does not guarantee acceptable service levels, so waiting time must also be evaluated using process capability analysis.
5. Continuous monitoring is required to maintain stability and identify any future changes in waiting time performance.

5. Process Capability Analysis

Process capability analysis is used to evaluate whether a process can consistently meet customer-defined quality requirements. It compares the natural variation of a process with specified acceptable limits, known as the Upper Specification Limit (USL) and Lower Specification Limit (LSL). The capability index **C_p** measures the potential capability of a process assuming it is perfectly centered, while **C_{pk}** measures the actual capability by considering both process variation and centering. In service processes such as banking, process capability analysis helps determine whether customer waiting times are consistently within acceptable limits and whether the process performance aligns with customer expectations.

5.1 Calculation of C_p (Process Potential Capability)

$$USL = 20, LSL = 0, \sigma = 5.76$$

$$C_p = \frac{20 - 0}{6 \times 5.76}$$

$$C_p = \frac{20}{34.56}$$

$$C_p = 0.58$$

Interpretation

Since $C_p < 1$, the process does not have sufficient potential capability to meet the specification limits.

5.2 Calculation of C_{pk} (Process Actual Capability)

$$\begin{aligned}
C_{pk} &= \min \left(\frac{USL - \bar{X}}{3\sigma}, \frac{\bar{X} - LSL}{3\sigma} \right) \\
&= \min \left(\frac{20 - 13.113}{3 \times 5.76}, \frac{13.113 - 0}{3 \times 5.76} \right) \\
&= \min \left(\frac{6.887}{17.28}, \frac{13.113}{17.28} \right) \\
&= \min (0.40, 0.76) \\
\boxed{C_{pk} = 0.40}
\end{aligned}$$

Interpretation

Since $C_{pk} < 1$, the process is not capable of consistently meeting the specification limits. The process mean is not well centered within the acceptable range.

5.3 Interpretation of Process Capability (Cp and Cpk)

1. The calculated values $C_p = 0.58$ and $C_{pk} = 0.40$ are both less than 1, indicating that the bank service process is not capable of consistently meeting the specified waiting time limits.
2. The C_p value below 1 shows that the natural variation in customer waiting time is wider than the acceptable range, meaning the process lacks potential capability even if it were perfectly centered.
3. Since C_{pk} is lower than C_p , the waiting time process is not properly centered within the specification limits, increasing the likelihood of waiting times exceeding acceptable standards.
4. Although the control charts indicate that the process is statistically stable, stability does not guarantee that service performance meets customer expectations.
5. The results indicate that improvements are required to reduce waiting time variation and improve service efficiency to consistently meet acceptable waiting time standards.

6. Quality Perspective, Process Improvement, And Quality Management Systems

Quality improvement in service operations focuses on reducing variation, eliminating operational inefficiencies, and ensuring consistent performance that meets customer expectations. In banking, customer waiting time is a key indicator of service quality and operational effectiveness. A comprehensive quality perspective involves not only identifying and correcting existing problems but also establishing systems that prevent their recurrence. This requires coordinated use of process improvement strategies, Quality Control monitoring, Quality Assurance planning, sampling-based performance evaluation, and structured quality management supported by audits and ISO standards.

6.1 Important Process Improvement Suggestions

Based on the analysis of quality control tools, control charts, and process capability results, the following practical and high-impact improvement measures are recommended:

1. **Increase staffing during peak hours** to balance customer demand and service capacity.
2. **Implement demand forecasting using historical data** to predict high-traffic periods and schedule staff accordingly.
3. **Introduce separate service counters for quick transactions** such as deposits, withdrawals, and enquiries to reduce queue congestion.
4. **Digitize documentation and approval processes** to minimize manual paperwork and reduce transaction time.
5. **Improve reliability and speed of core banking systems** to prevent service interruptions caused by technical delays.
6. **Provide regular training to staff** to enhance service speed, accuracy, and consistency.
7. **Implement structured queue management systems** such as token prioritization and service segmentation.
8. **Continuously monitor waiting time using control charts** to detect abnormal variations and take corrective action immediately.

These improvements target both reduction in average waiting time and reduction in variability, which together enhance overall process capability.

6.2 Role of Quality Control in the Bank Service Process

Quality Control focuses on monitoring service performance and detecting deviations from expected standards.

In the waiting-time process, Quality Control involves:

1. Measuring customer waiting time at regular intervals.
2. Using $\bar{X}$ and R control charts to monitor process stability.
3. Identifying abnormal variations caused by staffing imbalance or system malfunction.
4. Taking corrective action when control limits are exceeded.
5. Maintaining performance records for continuous monitoring.

Quality Control ensures that actual service delivery remains within acceptable operational limits.

6.3 Role of Quality Assurance in the Bank Service Process

Quality Assurance focuses on preventing service defects through systematic process design and standardization.

In banking operations, Quality Assurance includes:

1. Establishing standard operating procedures for customer handling.
2. Planning staffing levels based on predicted customer demand.
3. Ensuring regular maintenance and performance checks of banking systems.
4. Training employees to follow standardized service methods.
5. Designing efficient queue management structures.

Quality Assurance prevents problems before they occur, ensuring long-term service consistency.

6.4 Role of Sampling in Quality-Related Decision Making

Monitoring every service transaction is impractical in high-volume environments. Sampling enables efficient performance evaluation using selected observations.

In waiting-time analysis, sampling is used to:

1. Collect waiting time data from selected customers at different time periods.
2. Form rational subgroups for control chart analysis.
3. Estimate average service performance without observing every transaction.
4. Detect service trends efficiently with minimal data collection effort.

Sampling supports timely and cost-effective decision-making.

6.5 Quality Audits in the Bank Service Process

Quality audits are systematic reviews of operations to verify compliance with established standards and identify improvement opportunities.

In bank service queues, audits help to:

1. Evaluate adherence to service procedures.
2. Identify inefficiencies in staffing and queue management.
3. Verify accuracy of waiting-time monitoring systems.
4. Ensure corrective actions are properly implemented.
5. Promote accountability and continuous improvement.

Regular audits help sustain service quality over time.

6.6 Role of ISO 9001 in Maintaining Service Quality

ISO 9001 is an internationally recognized quality management standard that promotes process consistency and continuous improvement.

In banking service operations, ISO 9001 supports quality by:

1. Standardizing customer service procedures.
2. Establishing measurable performance indicators such as waiting time.
3. Encouraging risk-based operational planning.
4. Requiring documentation of processes and corrective actions “
5. Supporting continuous monitoring and systematic improvement.

Implementation of ISO 9001 ensures that quality management becomes structured, measurable, and sustainable.

7. Conclusion and learning outcomes

The conclusion of a Statistical Quality Control study summarizes the overall findings of the analysis and highlights the effectiveness of quality tools in understanding and improving real-life processes. Learning outcomes reflect the knowledge and skills gained through the application of statistical techniques, data interpretation, and process evaluation. In service-oriented organizations such as banks, concluding observations emphasize how systematic quality analysis supports operational efficiency, customer satisfaction, and continuous improvement.

Interpretation and Learning Outcomes

1. The project successfully applied Statistical Quality Control concepts to a real-life banking service process.
2. Customer waiting time was identified as a critical quality characteristic directly affecting service quality and customer satisfaction.

3. Quality control tools helped in systematically identifying major causes of excessive waiting time.
 4. Control charts demonstrated the importance of monitoring both process average and process variability.
 5. The study highlighted that a process can be statistically stable yet not capable of meeting customer expectations.
 6. Process capability analysis provided quantitative evidence for the need for improvement.
 7. The project improved understanding of data-driven decision-making in service operations.
 8. Practical insights were gained into applying theoretical quality concepts in real-world scenarios.
 9. The study reinforced the role of continuous monitoring and corrective action in quality management.
 10. Overall, the project enhanced analytical thinking and practical understanding of Statistical Quality Control applications.
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References

-  **Fishbone Diagram:** [Word Doc](#)
-  **Data & QC Charts:** [Excel File](#)
-  **Power BI ETL, Data Cleaning:** [Power BI File](#)
-  **Kaggle Dataset:** [Link](#)